Q1 • Single Correct •

Notes Bookmark Report

The work function of three photosensitive materials used to build photoelectric devices are given as : Sodium $(2.75\ \text{eV})$, copper $(4.65\ \text{eV})$ and gold $(5.1\ \text{eV})$. Which of the following statements is correct. (The frequency of visible light lies in the range 4×10^{14} Hz to 8×10^{14} Hz) ?

A Devices built by copper and gold can operate with visible light

B Devices built using sodium can operate with ultraviolet light

C All the devices can operate with infrared light

D All the devices can operate with visible light

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Q2 • Single Correct

Notes Bookmark Report

If $E =$ energy, $G =$ gravitational constant, $I =$ impulse and $M =$ mass, then dimensions of $\frac{GIM^2}{E^2}$ are same as that of.

A Time

B Mass

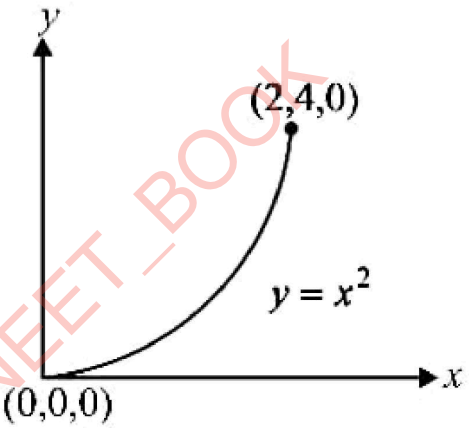
C Length

D Force

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Notes Bookmark Report

A force $\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$ is applied to a particle. The work done by the force when the particle moves from point $(0,\ 0,\ 0)$ to point $(2,\ 4,\ 0)$ as shown in the figure is



A $\frac{280}{5}$ J

B $\frac{140}{5}$ J

C $\frac{232}{5}$ J

D $\frac{192}{5}$ J

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Q4 • Single Correct •

Notes Bookmark Report

A soap bubble (surface tension $= T$) is charged to a maximum surface density of charge $= \sigma$. When it is just going to burst. Its radius R is given by:

A $R = \frac{\sigma^2}{8\varepsilon_0 T}$

B $R = 8\varepsilon_0 \frac{T}{\sigma^2}$

C $\mathrm{R} = \frac{\sigma}{\sqrt{8\varepsilon_0 \mathrm{T}}}$

D $R = \frac{\sqrt{8\varepsilon_0 \mathrm{T}}}{\sigma}$

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Q5 • Single Correct

Notes Bookmark Report

A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ is,

A 1/2

B 1/4

C 4

D 1

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Q6 • Single Correct •

Notes Bookmark Report

The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in second and ϕ_0 is a constant, the output voltage across the secondary coil is equal to 60 k V, then k is----

A 5

B 4

C 3

D 2

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Q7 • Single Correct

Notes Bookmark Report

A solid sphere of mass M and radius R has a spherical cavity of radius $\frac{R}{2}$ such that the centre of cavity is at distance R/ 2 from the centre of the sphere. A point mass m is placed inside the cavity at a distance R/4 from the centre of sphere. The gravitational pull between the sphere and the point mass m is:

A $\frac{11GMm}{R^2}$

B $\frac{14GMm}{R^2}$

C $\frac{GMm}{2R^2}$

D $\frac{GMm}{R^2}$

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Q8 • Single Correct

Notes Bookmark Report

Statement - 1: Two tuning forks having frequency 410 Hz and 524 Hz are kept close and made to vibrate. Beats will not be heard.
Statement - 2: Sound waves superimpose only when the frequencies of superposing waves are equal or nearly equal.

A. Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1.

B. Statement - 1 is True, Statement - 2 is True; Statement - 2 is NOT a correct explanation for Statement - 1.

C. Statement - 1 is True, Statement - 2 is False.

D. Statement - 1 is False, Statement - 2 is True.

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Q9 • Single Correct •

Notes Bookmark Report

A dipole of dipole moment $\vec{p} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at the point $A(2, -3, 1)$. The electric potential due to this dipole at the point $B(4, -1, 0)$ is equal to (All the parameters specified here are in SI units.),

A 2×10^9 V

B -2×10^9 V

C 3×10^9 V

D -3×10^9 V

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Q10 • Single Correct •

Notes Bookmark Report

The temperature of 5 moles of a gas at constant volume is changed from $100^{\circ}C$ to $120^{\circ}C$. The change in internal energy is 80 J. The total heat capacity of the gas at constant volume will be in J/K is :

A 8

B 4

C 0.8

D 0.4

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Q11 • Single Correct •

Notes Bookmark Report

The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is E. Let λ_1 be the de-Broglie wavelength of the proton and λ_2 be the wavelength of the photon. The ratio $\frac{\lambda_1}{\lambda_2}$ is proportional to?

A E^0

B $E^{\frac{1}{2}}$

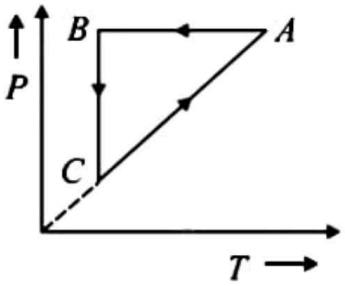
C E^{-1}

D E^{-2}

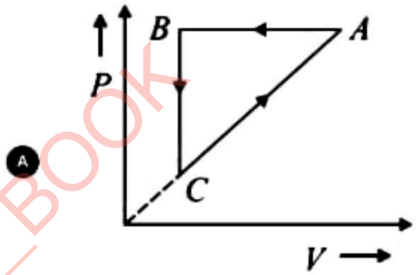
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Notes Bookmark Report

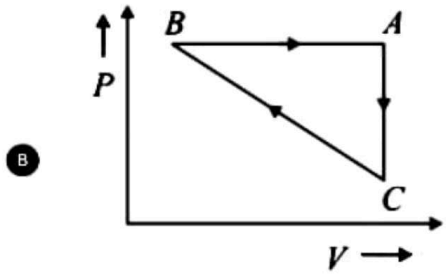
A cyclic process $ABCA$ is shown in $P-T$ diagram. What would be its $P-V$ graph:



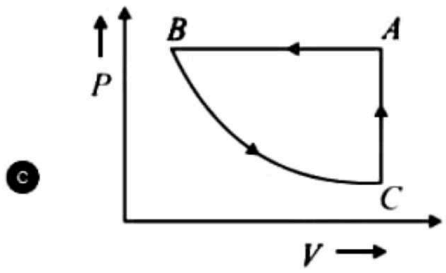
A



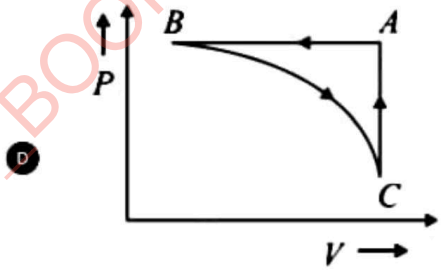
B



C



D

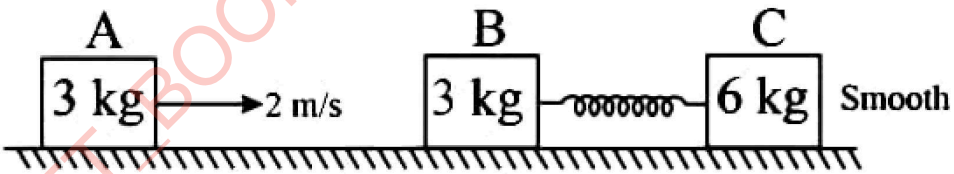


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Q13 • Single Correct •

Notes Bookmark Report

For the shown situation, if a collision between blocks A and B is perfectly elastic, then find the maximum energy (in joules) stored in the spring.



A 4

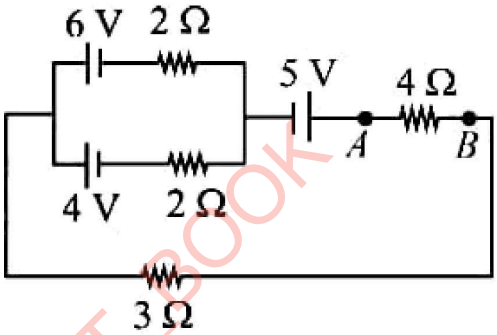
B 2

C 1

D 3

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In the circuit shown (figure), calculate the current through 6 V battery.



A (1/4) A

B (1/8) A

C (1/2) A

D None of these

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Q15 • Single Correct •

Notes Bookmark Report

A diffraction pattern is obtained using a beam of red light. What happens if the red light is replaced by blue light?

A Diffraction bands become narrower and crowded together.

B Diffraction bands become broader and farther apart.

C Bands disappear

D Diffraction band firstly become narrower and then become broader.

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Q16 • Single Correct

Notes Bookmark Report

A screw gauge gives the following reading when used to measure the diameter of a wire.
Main scale reading : 0 mm
Circular scale reading: 52 divisions
Given that 1 mm on the main scale corresponds to 100 divisions of the circular scale.
The diameter of the wire from the above data is

A 0.052 cm

B 0.026 cm

C 0.005 cm

D 0.52 cm

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The space has electromagnetic field which varies with time whose variation is given as:

$$\vec{B} = \begin{cases} -B_0\hat{k} & \text{if} \quad 0 \le t \le \frac{\pi m}{qB_0} \\ -B_0\hat{j} & \text{if} \quad \frac{\pi m}{qB_0} \le t \le \frac{2\pi m}{qB_0} \\ 0 & \text{if} \quad \frac{2\pi m}{qB_0} \le t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if} \quad 0 \le t \le \frac{2\pi m}{qB_0} \\ -E_0\hat{k} & \text{if} \quad \frac{2\pi m}{qB_0} \le t \le \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0\,\hat{i}$ at origin at $t = 0$ sec.

The coordinate of point on xy plane when it again passes through xy plane for the first time is of the form $\left(\frac{xmv_0}{q}\sqrt{\frac{v_0}{E_0\,B_0}},\ y\ \frac{mv_0}{qB_0},\ 0\right)$.

Find $x + y$?

A 5

B 6

C 3

D 4

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Q18 • Single Correct •

Notes Bookmark Report

A wheel of moment of inertia 2.5 Kg-m^2 has an initial angular velocity of 40 rads $^{-1}$. A constant torque of 10 Nm acts on the wheel. The time during which the wheel is accelerated to 60 rads^{-1} is.....

A 6 s

B 5 s

C 4.5 s

D 4 s

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Q19 • Single Correct •

Notes Bookmark Report

A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water +kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2. The apparent shift in the position of the bottom of the beaker when viewed from above is?

A $\left(1-\frac{1}{\mu_1}\right)h_2+\left(1-\frac{1}{\mu_2}\right)h_1$

B $\left(1+\frac{1}{\mu_1}\right)h_1+\left(1+\frac{1}{\mu_2}\right)h_2$

C $\left(1-\frac{1}{\mu_1}\right)h_1+\left(1-\frac{1}{\mu_2}\right)h_2$

D $\left(1+\frac{1}{\mu_1}\right)h_2-\left(1+\frac{1}{\mu_2}\right)h_1$

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Q20 • Single Correct •

Notes Bookmark Report

The escape velocity of an object from a planet is $16 \mathrm{kms}^{-1}$. If the escape velocity of the object from another planet having twice the density and three times the radius of the planet is $v\sqrt{2}\ \mathrm{ms}^{-1}$, then the value of v is____

A 12

B 48

C 18

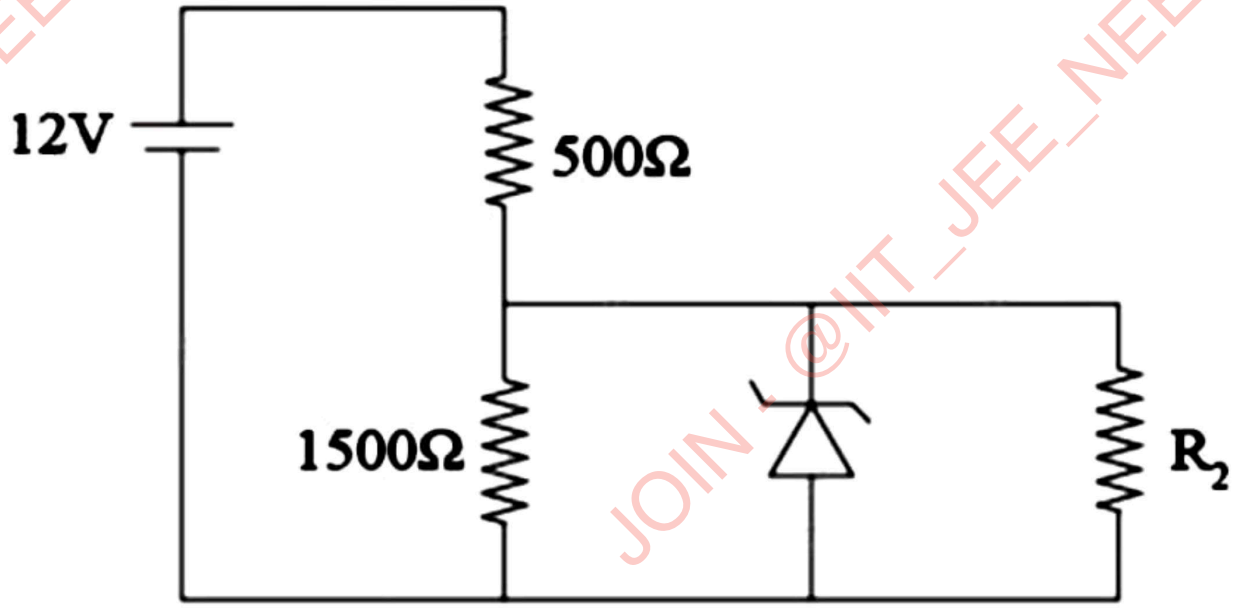
D 36

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Q21 • Numerical •

Notes Bookmark Report

A Zener diode of Zener break-down voltage 10 V is connected as shown in the figure. Current through Zener diode is



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Q22 • Numerical •

Notes Bookmark Report

A solenoid of length 60 cm with 15 turns per cm and area of cross section 4×10^{-3} m^2 completely surrounds another co-axial solenoid of same length and area of cross-section 2×10^{-3} m^2 with 40 turns per cm. Mutual inductance of the system is _____ mH. (Nearest Integer)

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Q23 • Numerical

Notes Bookmark Report

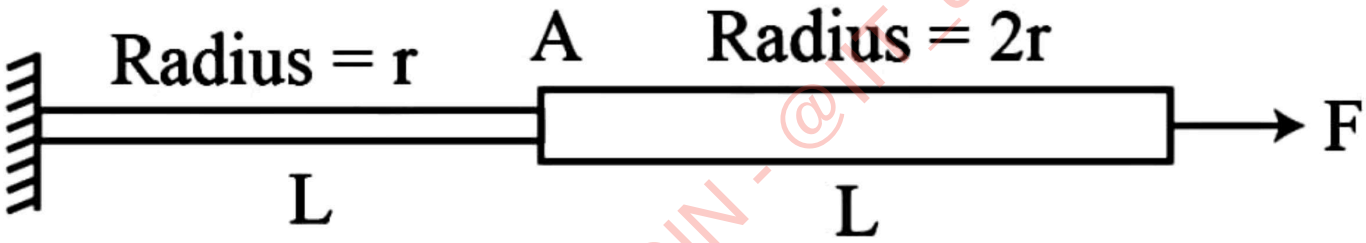
Position (in m) of a particle moving on a straight line varies with time (in sec) as $x = \frac{t^3}{3} - 3t^2 + 8t + 4(m)$. Consider the motion of the particle from $t = 0$ to $t = 5$ sec. S_1 is the total distance travelled and S_2 is the distance travelled during retardation. If $\frac{S_1}{S_2} = \frac{(3\alpha+2)}{11}$ then find α.

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Q24 • Numerical •

Notes Bookmark Report

Two steel wires of same length but radii r and $2r$ are connected together end to end and tied to a wall as shown. The force stretches the combination by 10 mm. How far does the midpoint A move. (in mm)

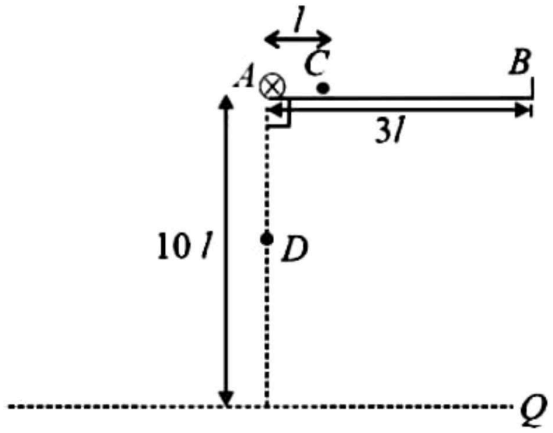


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Q25 • Numerical •

Notes Bookmark Report

A uniform thin rod of mass 'm' and length $\sqrt{3}l$ is released from rest from horizontal position as shown. When it passes through the vertical line AD, it gets broken at point 'C' due to some reason. Find the angle in radian rotated by rod BC till. Its centre of mass passes through the horizontal line PQ from the instant of breakage. $\left(g = 10\ m/s^2\right)$ (hinge is smooth and there is no thrust on one part of rod due to the other part at the time of breaking.)



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Notes Bookmark Report

The increasing order of Ag^+ ion concentration in
I. Saturated solution of AgCl
II. Saturated solution of AgI
III. 1M $Ag(NH_3)_2^+$ in 0.1M NH_3
IV. 1M $Ag(CN)_2^-$ in 0.1M KCN
Given:
K_{sp} of $AgCl = 1.0 \times 10^{-10}$
K_{sp} of $AgI = 1.0 \times 10^{-16}$
K_d of $Ag(NH_3)_2^+ = 1.0 \times 10^{-8}$
K_d of $Ag(CN)_2^- = 1.0 \times 10^{-21}$

A $I < II < III < IV$

B $IV < III < II < I$

C $IV < II < III < I$

D $IV < II < I < III$

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Q27 • Single Correct •

Notes Bookmark Report

The electronic configurations for some neutral atoms are given below
(I) $1s^2,\ 2s^2\ 2p^6,\ 3s^2$
(II) $1s^2,\ 2s^2\ 2p^6,\ 3s^1$
(III) $1s^2,\ 2s^2\ 2p^6,\ 3s^2\ 3p^2$
(IV) $1s^2,\ 2s^2\ 2p^6,\ 3s^2\ 3p^3$
Which of these is expected to have the highest second ionisation enthalpy?

A I

B II

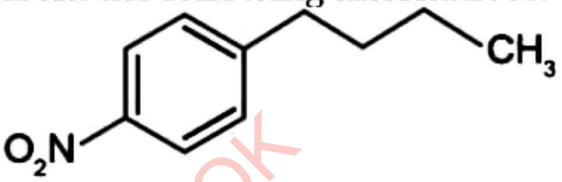
C III

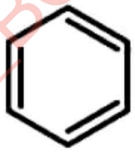
D IV

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Notes Bookmark Report

Identify the correct method for the synthesis of the compound shown below from the following alternatives.



A 

B 

C 

D 

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Q29 • Single Correct

Notes Bookmark Report

In the following sub questions, choose the correct answer from among the following possibilities and select correct code of your answer (Answer of 1, 2, 3 and 4 respectively)

1) The most stable low valent halide (1) $GeCl_2$ (2) $SnCl_2$ (3) $PbCl_2$.

2) A non existing halide (1) $SnCl_4$ (2) $PbCl_4$ (3) PbI_4.

3) A purely acidic oxide (1) PbO_2 (2) SnO_2 (3) SiO_2

4) Thermally most stable hydride (1) NH_3 (2) PH_3 (3) AsH_3.

A 3, 2, 1, 3

B 1, 3, 3, 1

C 3, 3, 3, 1

D 1, 1, 1, 3

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Q30 • Single Correct •

Notes Bookmark Report

Which of the following is a correct order of bond order ?

A $C_2 < NO < He_2^+ < O_2^-$

B $NO < C_2 < O_2^- < He_2^+$

C $He_2^+ < O_2^- < NO < C_2$

D $He_2^+ < O_2^- < C_2 < NO$

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Q31 • Single Correct

Notes Bookmark Report

A sample of hydrogen atoms de-excites from 6 th excited state to the ground state in one or more electronic transitions. If no spectral line is obtained in the Paschen and Brackett series, then the maximum number of spectral lines of different photon energies obtained will be -

A 21

B 15

C 10

D 7

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Q32 • Single Correct •

Notes Bookmark Report

The correct statement is .

A BI_3 is the weakest Lewis acid among the boron halides

B there is minimum $p\pi - p\pi$ back bonding in BF_3

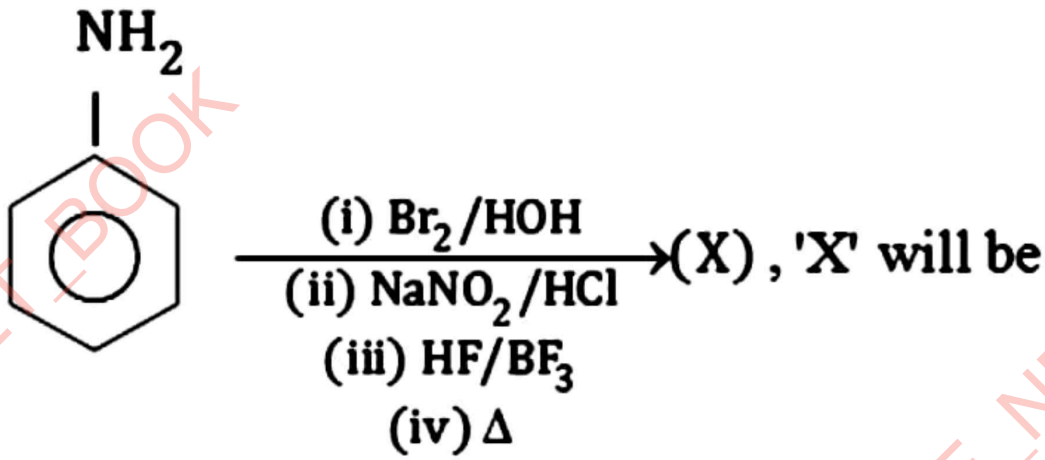
C BF_3 is the strongest Lewis acid among the other boron halides

D there is maximum $p\pi - p\pi$ back bonding in BF_3

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Notes Bookmark Report

In the reaction



A p-Bromofluorobenzene

B 2, 4, 6-Tribromofluorobenzene

C p-Bromoaniline

D 1. 3, 5-Tribromobenzene

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Notes Bookmark Report

Match list I with list II and select the correct answer using the codes given below the lists.

List I (Pair of isomers)	List II (Type of isomerism)
(p) (I) $[Co(NH_3)_6][Cr(CN)_6]$ (II) $[Cr(NH_3)_6][Co(CN)_6]$	1. Ionization
(q) (III) $[PrCl_2(NH_3)_4]Br_2$ (IV) $[PrBr_2(NH_3)_4]Cl_2$	2. Hydrate
(r) (v) $[Co(SCN)(NH_3)_5]Cl_2$ (vI) $[Co(NCS)(NH_3)_5]Cl_2$	3. Coordination
(s) (VII) $[Cr_2(H_2O)_6]Cl_3]$ (viII) $[CrCl_2(H_2O)_4]Cl.2H_2O$	4. Geometrical
	5. Linkage isomerism

(A)

p	q	r	s
4	1	5	2

(B)

p	q	r	s
1	3	2	5

(C)

p	q	r	s
3	1	5	2

(D)

p	q	r	s
1	3	5	2

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Q35 • Single Correct

Notes Bookmark Report

Among the following molecules/ions, what is the correct order of increasing s-character (in percentage) in the hybrid orbitals?

(I) CO_3^{2-} (II) XeF_4 (III) I_3^- (IV) NCl_3 (V) $BeCl_2$

A $II < III < IV < I < V$

B $II < IV < III < V < I$

C $III < II < I < V < IV$

D $II < IV < III < I < V$

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Q36 • Single Correct

Notes Bookmark Report

Identify the set of acidic oxides.

A Na_2O, CaO, BaO

B ZnO, PbO, BeO

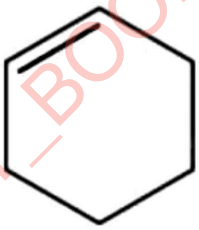
C CO, NO, N_2O

D Mn_2O_7, CrO_3, V_2O_5

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Notes Bookmark Report

If enthalpy of hydrogenation of $C_6H_6(l)$ into $C_6H_{12}(l)$ is -205 kJ and resonance energy of $C_6H_6(l)$ is -152 kJ mol^{-1} then enthalpy of hydrogenation of



(l) to $C_6H_{12}(l)$ is?

[Assume enthalpiies of vapourisation of all liquids involved are equal].

A -535.5 kJ/mol

B -238 kJ/mol

C -357 kJ/mol

D $-119kJ/mol$

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Q38 • Single Correct •

Notes Bookmark Report

Assertion: La (Z = 57) belongs to the d-block and is an element of group 3 of the periodic table.
Reason: All Lanthanoids belong to the group 3 of the periodic table.

A Both the assertion and reason statements are true and the reason is the correct explanation of the assertion

B Both assertion and reason statements are true but the reason is not the correct explanation of the assertion.

C The assertion statement is true but the reason statement is false

D Both the assertion and reason statements are false

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Q39 • Single Correct •

Notes Bookmark Report

For a complex reaction

$P \xrightarrow{k}$ Products

$E_{a_1} = 200\ kJ/mol$, $E_{a_2} = 90\ kJ/mol$, $E_{a_3} = 80\ kJ/mol$

Overall rate constant k is related to individual rate constant by the equation

$k = \left(\frac{k_1 k_2}{k_3}\right)^{2/3}$.

Activation energy (kJ/mol) for the overall reaction is

(mark answer to nearest integer)

A 240

B 160

C 210

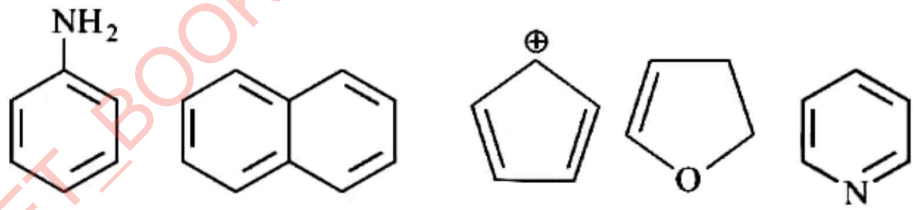
D 140

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Q40 • Single Correct •

Notes Bookmark Report

Total number of aromatic compounds from below is



A 1

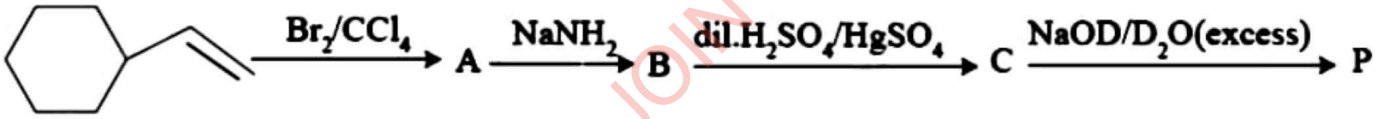
B 4

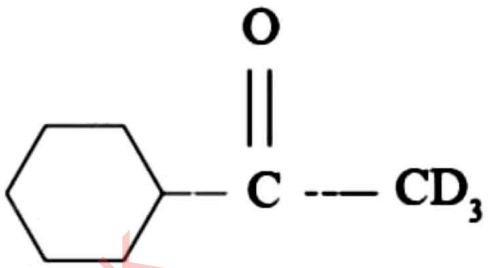
C 2

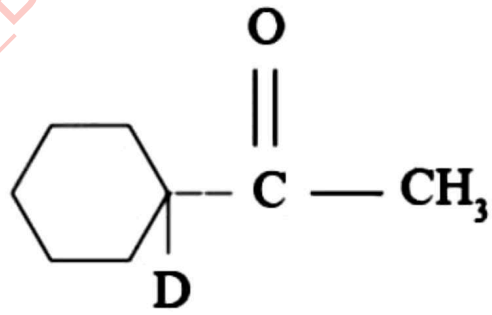
D 3

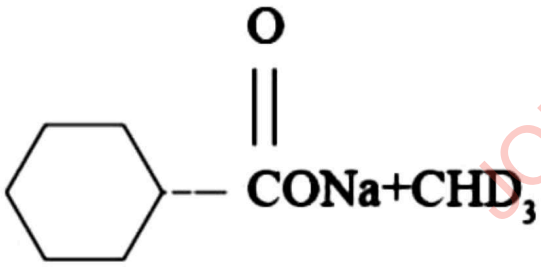
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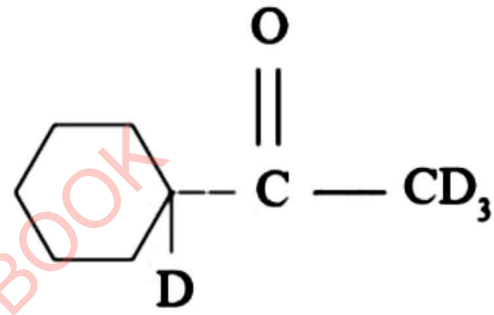
Final product (p) in the sequence of reaction is



(A) 

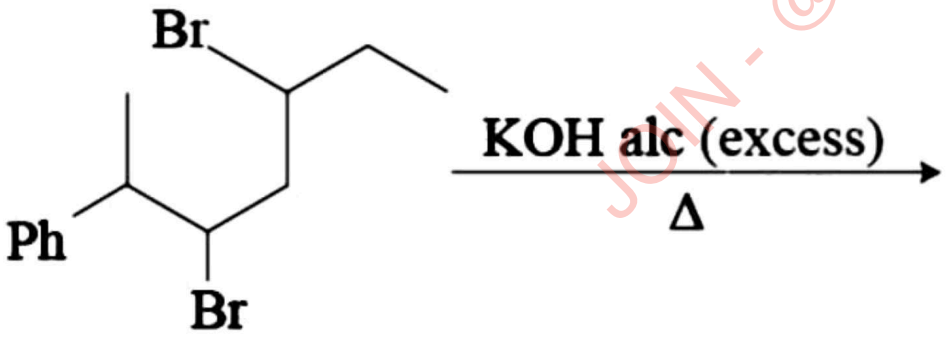
(B) 

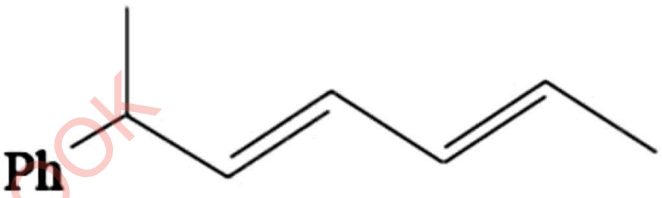
(C) 

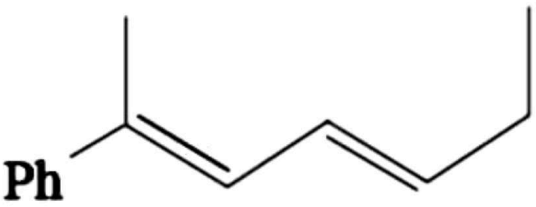
(D) 

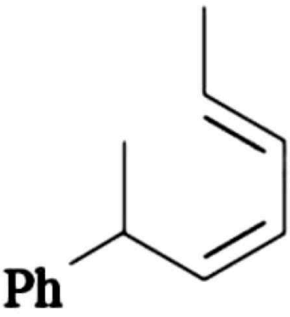
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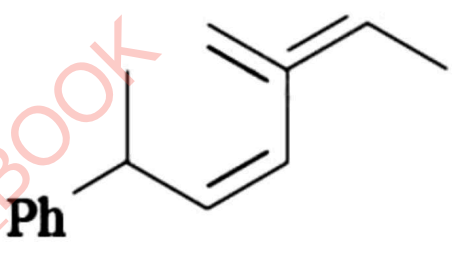
The major product of the following reaction is



(A) 

(B) 

(C) 

(D) 

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Q43 • Single Correct •

Notes Bookmark Report

Select the compound which cannot act as an oxidising agent and reducing agent?

A HNO_2

B H_2O_2

C $HClO_4$

D SO_2

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Q44 • Single Correct •

Notes Bookmark Report

The coordination number of Ni^{2+} is 4 .
$NiCl_2 + KCN$ (excess) $\rightarrow A$ (cyano complex)
$NiCl_2 +$ conc. HCl (excess) $\rightarrow B$ (chloro complex)
The IUPAC name of A and B are

A Potassium tetracyanonickelate (II), potassium tetrachloronickelate (II)

B Tetracyanopotassiumnickelate (II), tetrorachloropotassiumnickelate (II)

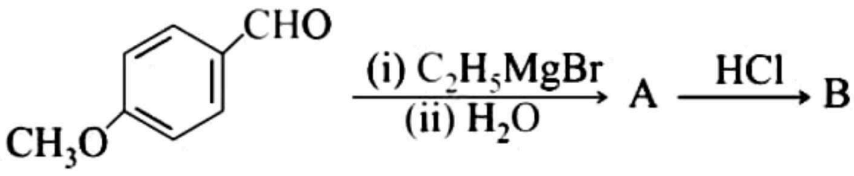
C Tetracyanonickel (II), tetrachloronickel (II)

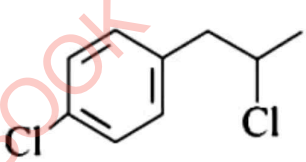
D Potassium tetracvanonickel (II), potassium tetrachloronickel (II)

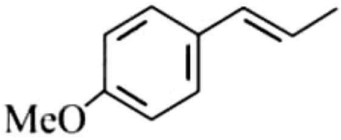
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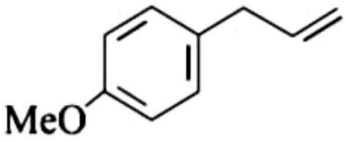
Notes Bookmark Report

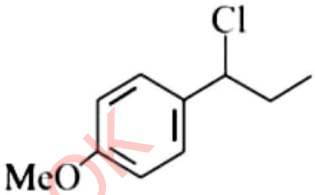
The major product B formed in the following reaction sequence is:



A 

B 

C 

D 

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Q46 • Numerical •

Notes Bookmark Report

1.5gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighting 1.46gm. The reaction responsible for loss of weight is :

$$Fe_2O_3(s) \rightarrow Fe_3O_4(s) + O_2(g)$$

What is the percentage by mass of Fe in original sample :-

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Q47 • Numerical

Notes Bookmark Report

Consider the following reversible reaction:
$NO + NO_3 \rightleftharpoons 2NO_2$
If 1.0mol of NO is mixed with 3.0mol of NO_3 , 'x' mole of NO_2 is produced at equilibrium. If 2.0mol of NO is added further, 'x' mol of NO_2 is further produced. What is the value of equilibrium constant?

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Q48 • Numerical

Notes Bookmark Report

A solution containing 8.0 g of nicotine in 92 g of water freezes 0.925 degrees below the normal freezing point of water. If the molal freezing point depression constant, $K_f = 1.85\,^{\circ}C\ mol^{-1}$, then the molar mass of nicotine is: (mark answer to nearest integer)

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Q49 • Numerical •

Notes Bookmark Report

For the following electrochemical cell at 298 K,

$Pt(s)|H_2(g, 1\ bar)|H^+(aq, 1M)||M^{4+}(aq), M^{2+}(aq)|Pt(s)$

$E_{cell} = 0.092V$ when $\frac{[M^{2+}(aq)]}{[M^{4+}(aq)]} = 10^x$

Given: $E^0_{M^{4+}|M^{2+}} = 0.151V; 2.303\ \frac{RT}{F} = 0.059\ V$

The value of x is (nearest integer).....

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Q50 • Numerical •

Notes Bookmark Report

There are some isomeric amines containing only one N atom. Each one of which forms a chloride salt on treatment with HCl containing 32.42% chlorine. None of them decolorizes Br_2 water.
How many of them does not give carbyl amine test?

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Q51 • Single Correct

Notes Bookmark Report

Let $f : R \to R$ be a function defined by $f(x) = -x^3 - 3x^2 - 6x + 1$. Number of integers in the solution set of x satisfying the inequality $f\left(f\left(x^3 + f(x)\right)\right) \geq f\left(f\left(-f(x) - x^3\right)\right)$ is?

A 3

B 4

C 5

D 6

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Q52 • Single Correct •

Notes Bookmark Report

The system of simultaneous equations $kx + 2y - z = 1, (k-1)y - 2z = 2$ and $(k+2)z = 3$ has a unique solution, if k equals

A -2

B -1

C 0

D 1

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Q53 • Single Correct •

Notes Bookmark Report

If $f(x) = \cos x - \int_0^x (x-t)f(t)dt$, then $f''(x) + f(x)$ is equal to

A $-\cos x$

B $-\sin x$

C $\int_0^x (x-t)f(t)dt$

D 0

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Q54 • Single Correct •

Notes Bookmark Report

If z is a complex number such that $z + i|z| = i\bar{z} + 1$, then $|z|$ is -

A 4

B 2

C 1

D 0

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Q55 • Single Correct

Notes Bookmark Report

If $\alpha_1, \beta_1, \gamma_1, \delta_1$ are the roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ and $\alpha_2, \beta_2, \gamma_2, \delta_2$ are the roots of the equation $ex^4 + dx^3 + cx^2 + bx + a = 0$ such that $0 < \alpha_1 < \beta_1 < \gamma_1 < \delta_1, 0 < \alpha_2 < \beta_2 < \gamma_2 < \delta_2$ $\alpha_1 - \delta_2 = 2 = \beta_1 - \gamma_2; \gamma_1 - \beta_2 = \delta_1 - \alpha_2 = 4$, then $a + b + c + d + e =$

A 10

B 12

C 6

D 8

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Q56 • Single Correct •

Notes Bookmark Report

If the area bounded by $f(x) = \tan^3 x + \tan x$ from $x = 0$ to $x = \frac{\pi}{4}$ is k square units, then the maximum value of $g(x) = k\sin x$ is $\left(\forall x \in \left[0, \frac{\pi}{4}\right]\right)$

A 2

B $\frac{1}{2\sqrt{2}}$

C 4

D 8

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Q57 • Single Correct •

Notes Bookmark Report

Let the function defined as $f(x) = \begin{cases} \frac{\sin[x^2]\pi}{x^2-3x-18} + ax^3 + b, & 0 \le x \le 1 \\ 2\cos\pi x + \tan^{-1}x, & 1 < x \le 2 \end{cases}$ is a differentiable function in the interval $(0, 2)$, then the value of $[a + b]$ equals, ($[\cdot]$ represents greatest integer function)

A 0

B -1

C -2

D 1

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Q58 • Single Correct

Notes Bookmark Report

4 cards are drawn from a well shuffled pack of 52 cards without replacement. If the fourth card is a king, then the chance that the first 3 cards are also king is $\frac{1}{20820+a}$, then a is equal to

A 4

B 1

C 2

D 5

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Q59 • Single Correct

Notes Bookmark Report

There are 3 identical black balls, 4 identical white balls and 2 identical red balls. The number of ways they can be arranged in a row so that at least one ball separates the balls of the same colour is

A 1210

B 1212

C 1244

D 1254

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Q60 • Single Correct •

Notes Bookmark Report

If the eccentricity of the hyperbola $x^2 - y^2\sec^2\theta = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\theta + y^2 = 25$, then smallest positive value of θ is $\frac{\pi}{p}$, where twice the value of ' p ' is

A 8

B 12

C 16

D 24

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Q61 • Single Correct •

Notes Bookmark Report

An equilateral triangle's sides increase at the rate of 2 cm/sec. If the area of its incircle increases at a rate of k cm^2/sec (when the length of the side is $\frac{6}{\pi}$ cm), then the value of k is

A 1

B 4

C 2

D 3

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Q62 • Single Correct •

Notes Bookmark Report

If $\sum_{i=1}^{5}(x_i - 6) = 5$ and $\sum_{i=1}^{5}(x_i - 6)^2 = 25$, then the standard deviation of observations $3x_1 + 2$, $3x_2 + 2$, $3x_3 + 2$, $3x_4 + 2$ and $3x_5 + 2$ is equal to

A 4

B 6

C 3

D 8

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Q63 • Single Correct •

Notes Bookmark Report

Consider the matrix $A = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}$ and B be a square matrix of order 2 such that $BB^T = B^TB = I$. Let $C = BAB^T$ and $D = [d_{ij}]_{2\times 2} = B^TC^6B$, then d_{11} is equal to

A 25

B 27

C 30

D 81

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Q64 • Single Correct •

Notes Bookmark Report

$A(1,0)$ and $B(0,1)$ and two fixed points on the circle $x^2+y^2=1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the triangle ABC is

A $x^2+y^2-2x-2y+1=0$

B $x^2+y^2-x-y=0$

C $x^2+y^2=4$

D $x^2+y^2+2x-2y+1=0$

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Q65 • Single Correct •

Notes Bookmark Report

If A and B are square matrices of order 3 x 3. where $|A| = 2$ and $|B| = 3$, then $\left|\left(A^{-1}\right)\text{adj}\left(B^{-1}\right) \cdot \text{adj}\left(2A^{-1}\right)\right| =$

A $\frac{8}{9}$

B $\frac{7}{9}$

C $\frac{2}{3}$

D $\frac{5}{3}$

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Q66 • Single Correct

Notes Bookmark Report

Let $f(x)=\left(\sin\left(\tan^{-1}x\right)+\sin\left(\cot^{-1}x\right)\right)^2-1, |x|<1$ If $\frac{dy}{dx}=\frac{1}{2}\frac{d}{dx}\left(\sin^{-1}(f(x))\right)$ and $y\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$, then $y\left(\frac{-1}{\sqrt{3}}\right)$ is equal to:

A $\frac{2\pi}{3}$

B $-\frac{\pi}{6}$

C $\frac{5\pi}{6}$

D $\frac{\pi}{3}$

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If $f(\theta) = \frac{1-\sin 2\theta + \cos 2\theta}{2\cos 2\theta}$ then value of $f(11^\circ) \cdot f(34^\circ)$ equals

A $\frac{1}{2}$

B $\frac{3}{4}$

C $\frac{1}{4}$

D 1

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Q68 • Single Correct •

Notes Bookmark Report

$A(27, -243, 81)$ is a point in space. B, C, D are images of A with respect to XY, YZ and ZX planes respectively. If the centroid of the triangle BCD is (α, β, γ), then $\alpha + \beta + \gamma =$

A -25

B 45

C 25

D -45

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Q69 • Single Correct •

Notes Bookmark Report

If $f(x)=\int 2^{\frac{\sin x}{\log_e 2}}\left(\frac{5\cos x+\cos^3 x}{(2-\sin x)\sqrt{3+\cos^2 x}}\right)dx$ and $f(0)=1$ then value of $f(\pi/2)$ equals

A $\sqrt{3}e$

B $\frac{\sqrt{3}e}{2}$

C $\frac{3e}{2}$

D $3e$

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Q70 • Single Correct

Notes Bookmark Report

Let g_n be the nth term of the geometric progression of positive numbers. If $\sum_{n=1}^{100} g_{2n} = \frac{10}{3}$ and $\sum_{n=1}^{100} g_{2n-1} = \frac{5}{9}$ then the common ratio of geometric progression, is

A 2

B 4

C 6

D 8

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Q71 • Numerical

Notes Bookmark Report

$$10 \int_0^{\pi} x\left(\sin^2(\sin x)+\cos^2(\cos x)\right) dx =$$

Round off ans to nearest integer (given $\pi^2 = 9.8$)

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Q72 • Numerical

Notes Bookmark Report

$OABC$ is a tetrahedron in with O as the origin and position vectors of points $A,\ B,\ C$ as $\hat{i}+2\hat{j}+3\hat{k},\ 2\hat{i}+\alpha\hat{j}+\hat{k}$ and $\hat{i}+3\hat{j}+2\hat{k}$ respectively, then the integral value of α to have shortest distance between $\overrightarrow{OA}$ & $\overrightarrow{BC}$ as $\sqrt{\frac{3}{2}}$, is

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Q73 • Numerical

Notes Bookmark Report

In a triangle ABC, $2\hat{i}+\hat{j}$, $5\hat{i}+\hat{j}$ & $2\hat{i}+7\hat{j}$ be the position vectors of points A, B & C respectively. Let $\overrightarrow{AD}$, $\overrightarrow{AB}$ & $\overrightarrow{AC}$ are non-coplanar vectors. If 'M' is minimum integral value of $\left(\overrightarrow{AD}\right)^2+\left(\overrightarrow{BD}\right)^2+\left(\overrightarrow{CD}\right)^2$ then sum of digits of M is

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Q74 • Numerical

Notes Bookmark Report

The constant term in the expansion of $\left(1+x+\frac{2}{x}\right)^{6}$ is

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Q75 • Numerical •

Notes Bookmark Report

If $y - \cos x \frac{dy}{dx} = y^2(1 - \sin x)\cos x, y(0) = 1$, then $y\left(\frac{\pi}{3}\right) =$

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